



DW_bin2gray

Binary to Gray Converter

Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History



Description

DW_bin2gray converts binary coded input *b* to Gray-coded output.

Figure 1-1

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
b	<i>width</i> bits	Input	Binary coded input data
g	<i>width</i> bits	Output	Gray coded output data

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 1	Input word length

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW_bin2gray_cfg_sim	Design unit name for VHDL simulation
dw/dw01/src/DW_bin2gray_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_bin2gray.v	Verilog simulation model source code

Reflected binary Gray code sequences can be constructed iteratively starting with the simplest two element sequence of 0 and 1 (see [Figure 1-2](#) on page 2). Each iteration doubles the sequence by concatenating the previous sequence with a reversed (reflected) copy of itself. In addition, a new Most Significant Bit (MSB) is added to each element with its value being 0 for the forward copy and 1 for the reflected copy.

Figure 1-2 Gray Code Number Relationship to Corresponding Binary Number

Basic 2-Element Sequence width = 1		4-Element Sequence width = 2		8-Element Sequence width = 3		16-Element Sequence width = 4	
Binary	Gray Code	Binary	Gray Code	Binary	Gray Code	Binary	Gray Code
0	0	0 0	0 0	0 0 0	0 0 0	0 0 0 0	0 0 0 0
1	1	0 1	0 1	0 0 1	0 0 1	0 0 0 1	0 0 0 1
		1 0	1 1	0 1 0	0 1 1	0 0 1 0	0 0 1 1
		1 1	1 0	0 1 1	0 1 0	0 0 1 1	0 0 1 0
				1 0 0	1 1 0	0 1 0 0	0 1 1 0
				1 0 1	1 1 1	0 1 0 1	0 1 1 1
				1 1 0	1 0 1	0 1 1 0	0 1 0 1
				1 1 1	1 0 0	0 1 1 1	0 1 0 0
						1 0 0 0	1 1 0 0
						1 0 0 1	1 1 0 1
						1 0 1 0	1 1 1 1
						1 0 1 1	1 1 1 0
						1 1 0 0	1 0 1 0
						1 1 0 1	1 0 1 1
						1 1 1 0	1 0 0 1
						1 1 1 1	1 0 0 0

Related Topics

- [Math - Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_arith.all;

entity DW_bin2gray_func is
    generic (func_width : positive := 8);
    port (func_b : in  std_logic_vector(func_width-1 downto 0);
          g_func : out std_logic_vector(func_width-1 downto 0));
end DW_bin2gray_func;

architecture func of DW_bin2gray_func is
begin
    -- function inference of DW_bin2gray
    g_func <= DWF_bin2gray (func_b);
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_bin2gray_func (func_b, g_func);

    parameter func_width = 8;

    input  [func_width-1 : 0] func_b;
    output [func_width-1 : 0] g_func;

    // pass "width" parameters to the inference functions
    parameter width = func_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW_bin2gray_function.inc"

    // function inference of DW_bin2gray
    assign g_func = DWF_bin2gray (func_b);
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_bin2gray_inst is
    generic (inst_width : positive := 8);
    port (inst_b : in  std_logic_vector(inst_width-1 downto 0);
          g_inst : out std_logic_vector(inst_width-1 downto 0));
end DW_bin2gray_inst;

architecture inst of DW_bin2gray_inst is
begin
    -- instance of DW_bin2gray
    U1 : DW_bin2gray
        generic map (width => inst_width)
        port map (b => inst_b,
                  g => g_inst);
end inst;

-- pragma translate_off
configuration DW_bin2gray_inst_cfg_inst of DW_bin2gray_inst is
    for inst
    end for;
end DW_bin2gray_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_bin2gray_inst (inst_b, g_inst);

    parameter inst_width = 8;

    input  [inst_width-1 : 0] inst_b;
    output [inst_width-1 : 0] g_inst;

    // Please add +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator
    // command line (for simulation).

    // instance of DW_bin2gray
    DW_bin2gray #(inst_width)
        U1 (.b(inst_b), .g(g_inst));
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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